Introduction of Machine Learning

Applied Machine Learning

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Contents

- Module Information
- Machine Learning Introduction
- Machine Learning Process
- Machine Learning Approaches

Teaching Team

(3 Academics, 2 Teaching Associates, 1 PGR Teaching Assistant)

Lecture



Yukun Lai



Yuhua Li
Module lead



Yipeng Qin

Practical



Yingying Wu
Lab lead



Suhas Devmane

Deviprabha Surendran

Module Organisation

Part 1

Basic Machine Learning concepts

+

Popular General ML methods

Part 2

Neural networks / Deep learning

NLP + Image processing

Module plan: Part 1

- Week 1: Introduction to machine learning

  Data pre-processing issues and techniques
- Week 2: Regression, Generalisation and model evaluation
- Week 3: Classification
- Week 4: Ensemble learning

Class test: in **Week 7**, check your timetable

Plan may subject to change

Assessment

[Class test, 50%]

Week 7

1.5 hours + 30 mins for upload

Computer timed online test.

[Group project, 50%]

Design and implement neural networks for solving a problem.

Write a group report.

Write an individual report.

Delivery method

- On campus in-person **lectures**
  - All materials on Learning Central
- On campus in-person practical **lab** sessions
  - See your timetable
- Project **supervision**

Check weekly map on LC

Your efforts

- Attend sessions
- Learning theory
- Do practice
- Self research



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Why Machine Learning

- Face authentication



- Driverless cars



- Computer game



- LLM

OpenAI

ChatGPT

- Recommendation

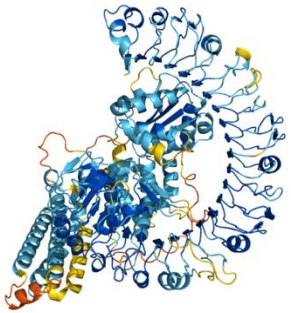


amazon



- AI for Science

AlphaFold

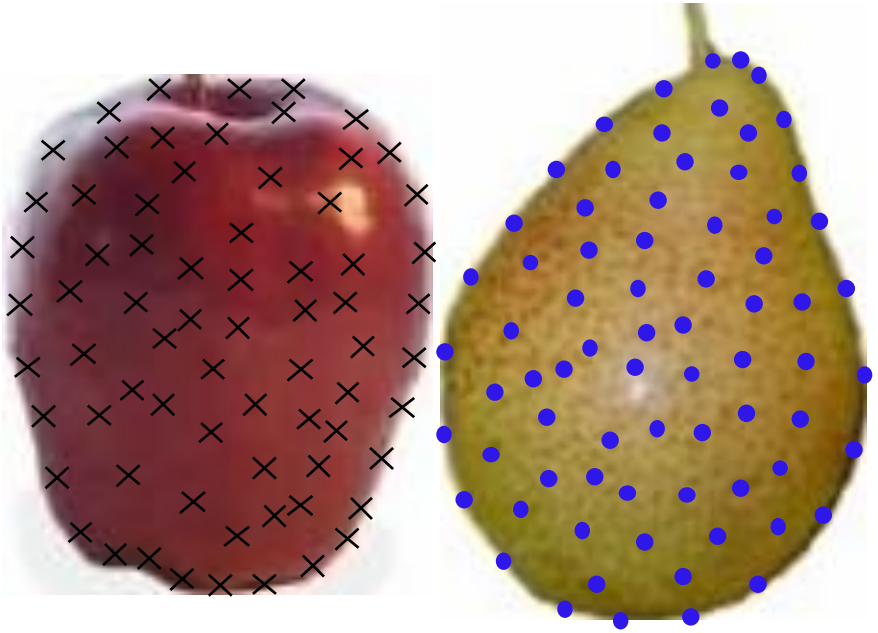


Machine Learning Application Examples

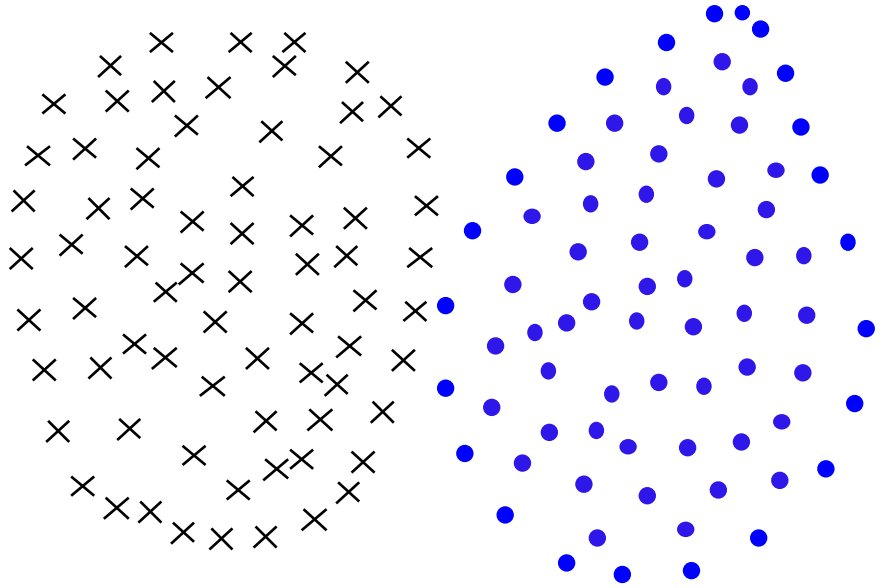
- Search engines use ML to improve search results based on users' search queries.
- Google Maps uses ML to suggest the fastest routes by analysing anonymous traffic data from smartphones.
- PayPal uses ML to combat fraud.
- Netflix uses ML for recommendation and customer retention.
- Uber uses ML to determine arrival times and pickup locations.
- Spotify uses ML to figure out your likes and dislikes and provides you with a list of related tracks.

Introduction

- Object Representation Using Samples/data (fictitious)



Introduction

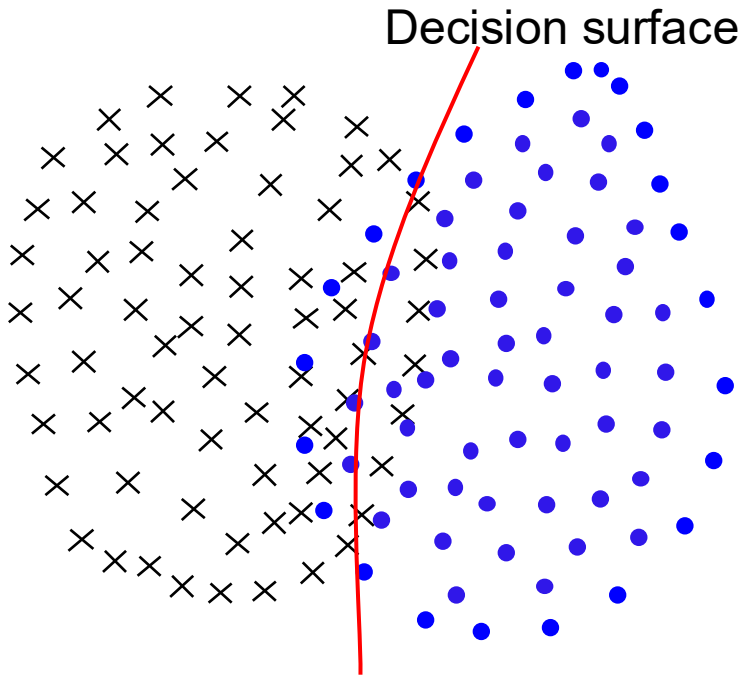


Iain Murray fruit dataset

Fruit type	mass	height	width	Label	3c label		
Granny smith apple	192	8.4	7.3	1	1		1 - apple
Granny smith apple	180	8	6.8	1	1		2 - orange
Granny smith apple	176	7.4	7.2	1	1		3 - lemon
Mandarin	86	6.2	4.7	2	2		
Mandarin	76	5.8	4	2	2		
Braeburn apple	178	7.1	7.8	3	1		
Braeburn apple	154	7	7.1	3	1		
Golden delicious apple	164	7.3	7.7	4	1		
Golden delicious apple	168	7.5	7.6	4	1		
Cripps Pink Washington USA apple	162	7.5	7.1	5	1		
Cripps Pink Washington USA apple	140	7.3	7.1	5	1		
Cripps Pink Washington USA apple	170	7.6	7.9	5	1		
Lane Late Spanish jumbo orange	342	9	9.4	6	2		
Lane Late Spanish jumbo orange	362	9.6	9.2	6	2		
Morrisons "selected seconds" orange	204	7.5	9.2	7	2		
Morrisons "selected seconds" orange	140	6.7	7.1	7	2		
Morrisons "selected seconds" orange	160	7	7.4	7	2		
Morrisons "selected seconds" orange	158	7.1	7.5	7	2		
Morrisons "selected seconds" orange	210	7.8	8	7	2		
Morrisons "selected seconds" orange	164	7.2	7	7	2		
Turkey Navel class 1 cal 6 orange	190	7.5	8.1	8	2		
Turkey Navel class 1 cal 6 orange	154	7.2	7.2	8	2		
Belsan Spanish lemons	194	7.2	10.3	9	3		
Belsan Spanish lemons	174	7.3	10.1	9	3		
lemons unknown type from local shop	132	5.8	8.7	10	3		
lemons unknown type from local shop	152	6.5	8.5	10	3		
lemons unknown type from local shop	118	6.1	8.1	10	3		

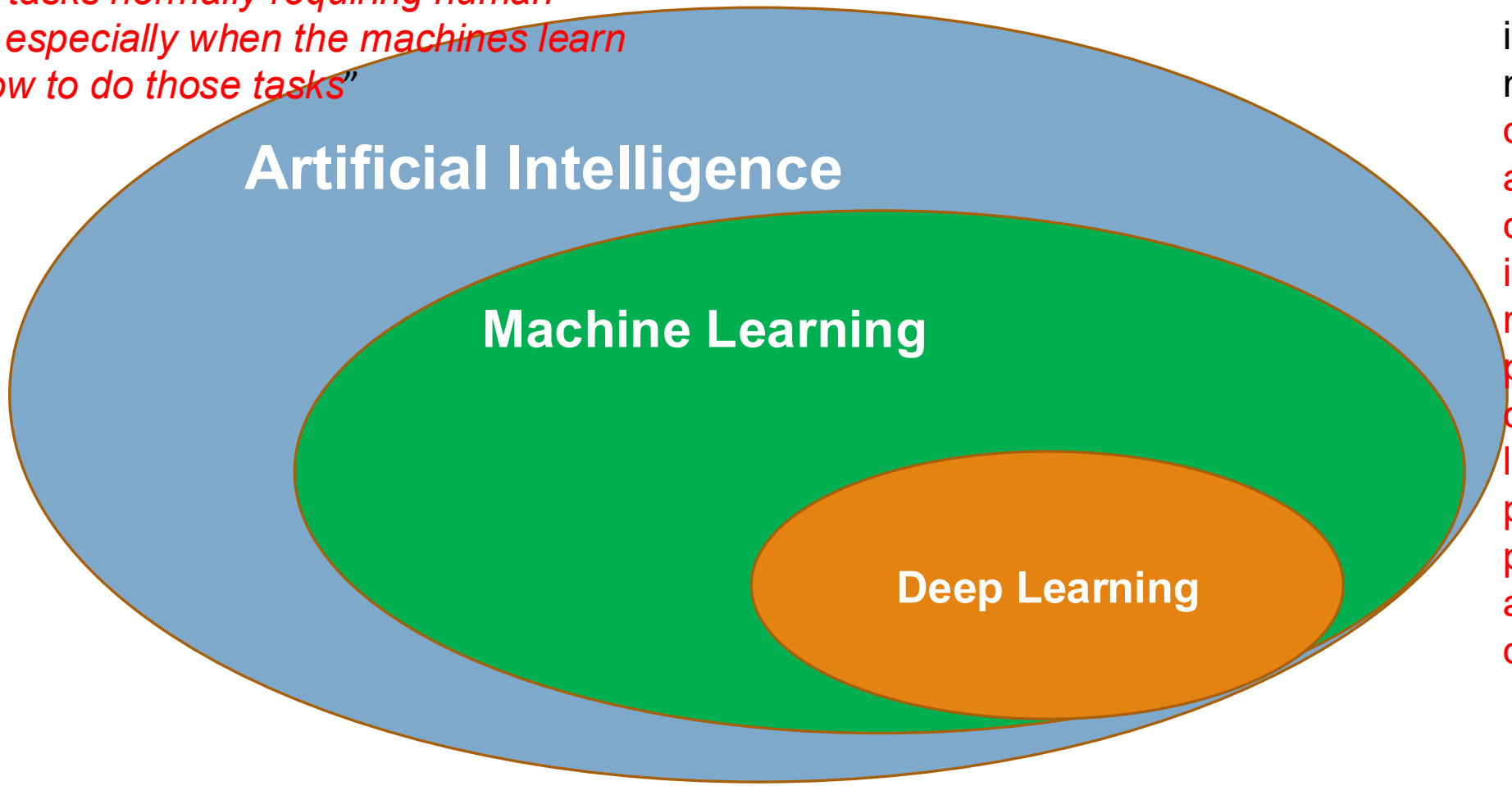
Tabular data

Introduction



AI, Machine Learning and Deep Learning

UK National AI Strategy definition: AI is "*Machines that perform tasks normally requiring human intelligence, especially when the machines learn from data how to do those tasks*"



UK legal definition: "artificial intelligence" means technology designed to approximate cognitive abilities including reasoning, perception, communication, learning, planning, problem solving, abstract thinking or decision making.

Machine Learning

- It is the ability of machines to learn and solve problems just like humans
- Algorithms that learn models to discover patterns and relationships or uncover "hidden insights" from available data for prediction, description and diagnosis
- “Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalise to unseen data, and thus perform tasks without explicit instructions.” (Wikipedia)

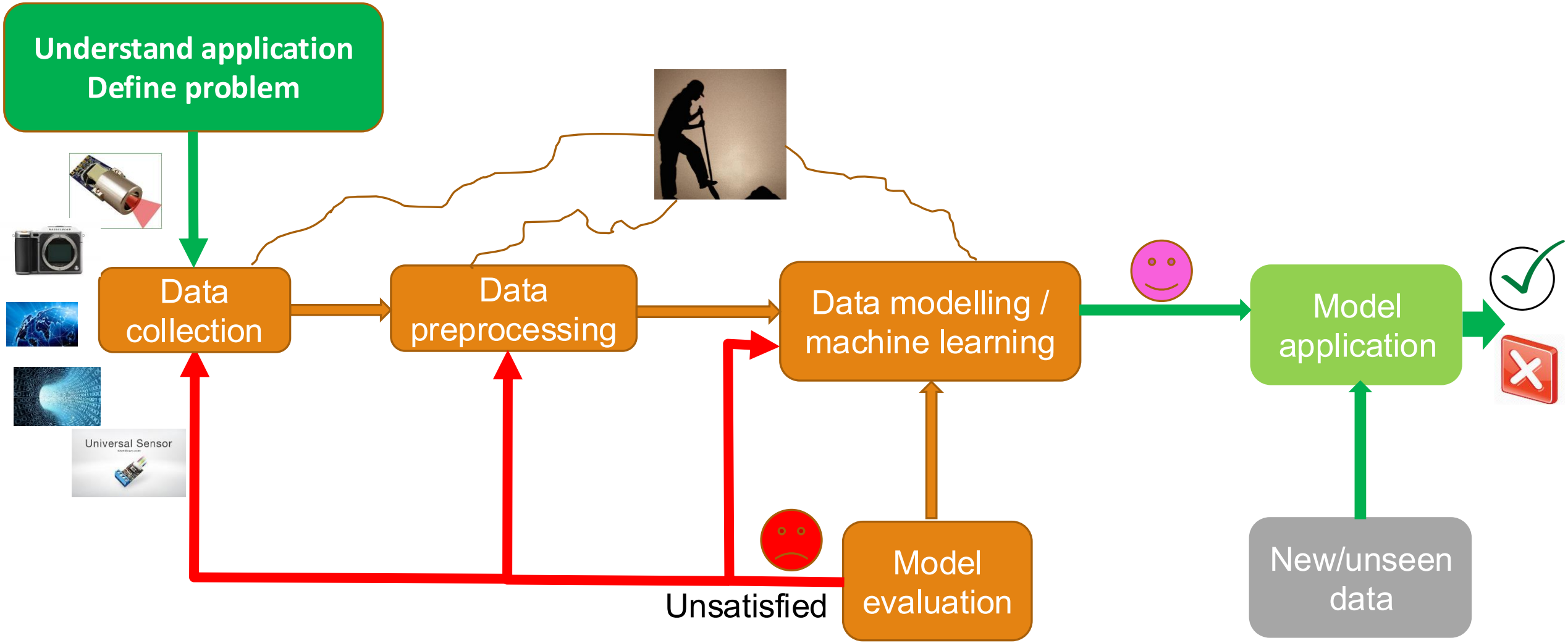


$$y = f(X)$$

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Logical Process of Machine Learning



Data Collection - Types of Data Sets

- Record
  - Relational records
  - Data matrix, e.g., numerical matrix, crosstabs
  - Document data: text documents, term-frequency vector
  - Transaction data
- Graph and network
  - World Wide Web
  - Social or information networks
  - Molecular Structures
- Ordered
  - Video data: sequence of images
  - Temporal data: time-series
  - Sequential Data: transaction sequences
  - Genetic sequence data
- Spatial, image and multimedia
  - Spatial data: maps
  - Image data: pictures
  - Video data: clips, films

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Important Characteristics of Structured Data

- Dimensionality
  - Curse of dimensionality
- Sparsity
  - Only presence counts
- Resolution
  - Patterns depend on the scale
- Distribution
  - Centrality and dispersion

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

Data Objects

- Data sets are made up of data objects.
- A **data object** represents an entity.
- Examples:
  - sales database: customers, store items, sales
  - medical database: patients, treatments
  - university database: students, professors, courses

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

- Also called ***points***, *samples , examples, instances, objects, tuples, vectors, patterns*.
- Data objects are described by **attributes**.
- Database
  - rows -> data objects; columns ->attributes.

Attributes

- **Attribute** (or **features**, **variables, dimensions, predictors**): a data field, representing a characteristic or feature of a data object.
  - *E.g., customer _ID, name, address*

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
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- Types:
  - Nominal
  - Binary
  - Ordinal

  } Categorical feature

  - Numeric: quantitative

Attribute Types

- **Nominal:** categories, states, or "names of things"
  - *Hair_color* = {*auburn, black, blond, brown, grey, red, white*}
  - marital status, occupation, ID numbers, post codes
- **Binary**
  - Nominal attribute with only 2 states (0 and 1)
  - Symmetric binary: both outcomes equally important
    - e.g., gender
  - Asymmetric binary: outcomes not equally important.
    - e.g., medical test (positive vs. negative)
    - Convention: assign 1 to most important outcome (e.g., HIV positive)
- **Ordinal**
  - Values have a meaningful order (ranking) but magnitude between successive values is not known.
  - *Size* = {*small, medium, large*}, grades, army rankings

Discrete vs. Continuous Attributes

- **Discrete Attribute**
  - Has only a finite or countably infinite set of values
    - E.g., zip codes, profession, or the set of words in a collection of documents
  - Sometimes, represented as integer variables
  - Note: Binary attributes are a special case of discrete attributes
- **Continuous Attribute**
  - Has real numbers as attribute values
    - E.g., temperature, height, or weight
  - Practically, real values can only be measured and represented using a finite number of digits
  - Continuous attributes are typically represented as floating-point variables

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Machine Learning Approaches

- Primary types of machine learning
  - Supervised learning
  - Unsupervised learning
  - Reinforcement learning

Machine Learning Approaches

- Supervised learning
  - The outputs/targets are predetermined
  - Oriented towards prediction and interpretation
  - Classification, e.g., decide if a tumour is benign or malignant
  - Regression, e.g., bitcoin price prediction

UCI Banknote Authentication (Wavelet Transformed image)

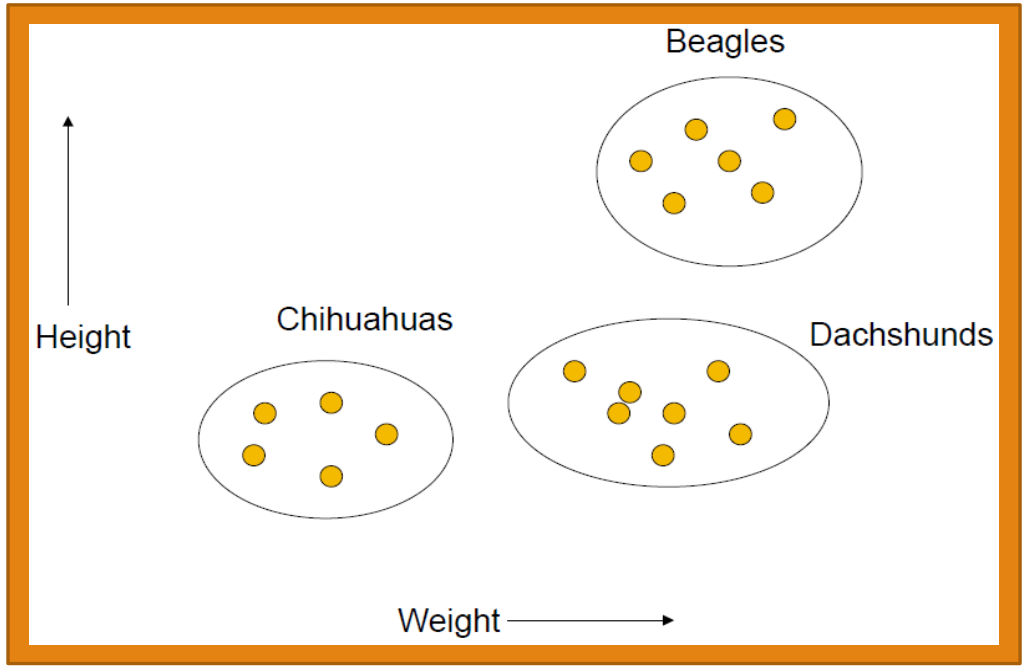
	variance	skewness	curtosis	entropy	class
Banknote 1	0.01727	8.693	1.3989	-3.9668	genuine
Banknote 2	3.2414	0.40971	1.4015	1.1952	genuine
Banknote 3	2.2504	3.5757	0.35273	0.2836	genuine
Banknote 4	-1.3971	3.3191	-1.3927	-1.9948	forged
Banknote 5	0.39012	-0.14279	-0.031994	0.35084	forged
Banknote 6	-1.6677	-7.1535	7.8929	0.96765	forged

Machine Learning Approaches

- Unsupervised learning
  - Not provided with classification labels. The basic task is to develop classification labels automatically
  - Clustering (e.g, K-means)

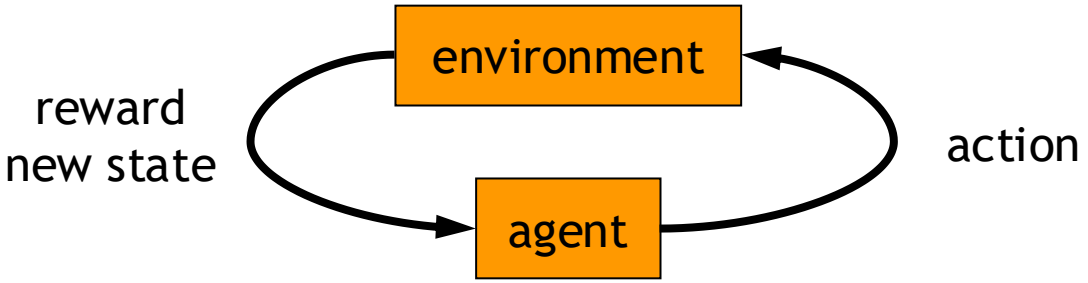
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Machine Learning Approaches

- Reinforcement learning
  - more general than supervised/unsupervised learning
  - learn from interaction with external environment to achieve a goal
  - Examples
    - Game playing: player knows whether it wins or loses, but not know how to move at each step
    - Control: a traffic system can measure the delay of cars, but not know how to decrease it.



Model Evaluation

- Evaluates the performance of a learned model to provide a confidence in future application of the model to unseen data
  - A separate section on this later

Summary

- Learning a model from data involves a systematic procedure.
- It is vital to invest time and effort on data pre-processing.
- The use of a machine learning algorithm depends on data characteristics and application domain.

Bibliography (books)

- Practical books
  - *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd Edition by Aurélien Géron, 2022
  - *Deep Learning with Python*, 2nd edition by Francois Chollet, 2021
- Academic books
  - *Pattern Recognition and Machine Learning*, by Christopher Bishop, 2007
    https://www.microsoft.com/en-us/research/people/cmbishop/prml-book/
  - *Deep Learning: Foundations and Concepts*, by Christopher Bishop, 2024
    https://link.springer.com/book/10.1007/978-3-031-45468-4
- More books
  - Deep Learning, by Ian Goodfellow, Yoshua Bengio, Aaron Courville, 2016
    https://www.deeplearningbook.org/
  - *Machine Learning: The Art and Science of Algorithms that Make Sense of Data*, by Peter Flach, 2012.
  - *Understanding Machine Learning: From Theory to Algorithms*, by By Shai Shalev-Shwartz and Shai Ben-David, 2014: http://www.cs.huji.ac.il/~shais/UnderstandingMachineLearning/
  - *Machine Learning for Engineers*, by Osvaldo Simeone, Cambridge University Press, 2022.
  - *Understanding Deep Learning*, by Simon J.D. Prince, MIT Press, 2023. https://udlbook.github.io/udlbook/
- References on some topics may also be provided in the Learning Central

https://schoolmanagement.cardiff.ac.uk/registers

Practical Information

Python libraries

Essentials for this module:

- **Sklearn** (scikit-learn) -> Machine Learning
- **Keras (Tensorflow)** -> High-level API for deep learning
- **Numpy** -> Mathematical functions, vectors

Very useful:

- **Pandas** -> Data analysis
- **Matplotlib** -> Visualization
- **Pytorch** -> Deep learning library
- **NLTK** -> Statistical text processing
- **Spacy** -> Natural language processing
- **Gensim** -> Vectors, topic modelling
- ……



Google Colab

- **Colaboratory** is a free Jupyter notebook environment that requires no setup and runs entirely in the cloud.
- It contains all main Python libraries required for this course.

Suggested for this module!

https://colab.research.google.com



Google Colab

- https://colab.research.google.com
- It follows the same block-based format as **Jupyter Notebooks**. Practical assignments will be delivered with this format.
- **Google Drive** is used as storage unit.
- It has free **GPU**-time which can be very useful for the second semester.

Colab: Some useful introductions/tutorials

- Introduction: https://colab.research.google.com/notebooks/welcome.ipynb#

- Google Colab tutorial - An introduction to Google Colab, How to upload files, How to connect to Google Drive and resume training.

- Gettings started (blog post): https://towardsdatascience.com/getting-started-with-google-colab-f2fff97f594c